

SPENCER DELCORE

Computer Engineering - University of Waterloo

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QUALIFICATIONS

Languages	Python, C, C++, C#, VHDL/Verilog, Java, MATLAB, PHP, JavaScript
Technologies	FPGAs, Micro-controllers, UNIX, Git, MySQL, PostgreSQL
Embedded Knowledge	Parallel/Serial Interfacing, Memory Management, SW/HW Debugging

WORK EXPERIENCE

Cognitive Systems Corp. - *Augmented Reality (AR) Developer, Co-op* Jan. 2017 - April. 2017
Quantum Valley Investments, Waterloo, ON

- Responsible for creating an AR demonstration to visualize wireless signals
- Dynamically generated interactive holograms using Unity and C#
- Create a Python program that interfaces with Cognitive's system on a chip
- Communicate between the SOC and the AR headset through websockets

Ingle International Inc. - *Software Engineering, Co-op* May. 2016 - Aug. 2016
Toronto, ON

- Accountable for building and maintaining version two of their API using PHP and the Laravel framework
- Built a dynamically themed Android app using Java and React Native
- Researched, presented, and participated in critical development meetings

VIZIYA - *Software Developer, Co-op* Sept. 2015 - Dec. 2015
Waterdown, ON

- Designed and developed quality ASP.NET and C# applications
- Performed SQL queries to retrieve and manipulate data from Oracle and SAP databases
- Generated user guides, training manuals, and documentation

Emerging Star Investment Group - *Software Developer, Co-op* Jan. 2015 - April 2015
Accelerator Centre, Waterloo, ON

- Developed and maintained financial web-based software using PHP and the Laravel framework
- Built scripts to gather, manipulate, and store data using Python

PROJECTS

VHDL Compiler - *Java, VHDL* 2017

- Created a VHDL synthesizer and simulator using Java
- Uses a subset of VHDL (combinational circuits) as the input language and produces circuit gate diagrams and simulations
- Gathered important compiler concepts such as regular languages, regex, DFA, lexing, recursive decent parsing, and context-free grammars

Room Sensors - *C++, Python, Bash, JavaScript* 2017

- Plots information stored in a database containing data from Arduino sensors
- Communicate and store sensor data from an Arduino to a Raspberry Pi using a created serial protocol
- Chart data from JSON reports received from a Raspberry Pi through websockets
- Created an install script that sets up the websocket, web page, cron job, Arduino, and MySQL database

DAC Audio Player - *Altera FPGA, C* 2016

- Implemented a 16-bit sampling stereo digital to analog audio player with push buttons to control playback
- Gained experience with Altera audio IP core, audio sampling and buffering, and interrupts
- Made use of the FatFS CLI to mount and interpret WAV files from a microSD card